Reflection on Cognitivism

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General Notes

- 1950s, move away from behavioural models that mainly focused on observable behaviour (what you can see a learner doing)
    - Greater focus on cognitive sciences
        - Looks at what happens in the mind when people learn
    - Includes complex mental processes like:
        - Thinking
        - Problem solving
        - Language
        - Forming concepts and categories
        - Processing information
- Move from **behavioral orientation**
    - Focusing on getting students to show the desired observable behaviour by changing or controlling the stimulus/environment
- To **cognitive orientation**
    - Focusing on helping students develop and improve their mental processing
        - I.e., using instructional procedures that guide how students think, process information, and interact with the instructional system

Understandings of Learning / Knowledge

- Learner is viewed as an **active participant** in learning
    - Not just receiving information
    - Actively thinking about it & working with it
- Learning is not just a change in observable behaviour (as per behaviorism)
    - Learning involves changes in how a person organizes, stores, and understands information in their mind
    - Mind is like **information processor** (e.g., computer)
        - Takes in information, processes it, organizes it, stores it, and retrieves it when needed
        - As learning happens, learner can move from one level of understanding/schema to another
- Knowledge is made of internal mental structures (schemas/schemata)
    - **Schema** = mental framework or “folder” that helps organize information and understand things
- Learning happens when a person successfully:
    - Takes in new information
    - Organizes it
    - Codes it (turns it into a form the mind can work with and remember)

    - Stores it
    - Connects it to their existing mental framework
- **Knowledge acquisition** is a mental activity
    - Learning means the learner has to mentally code, organize, and structure information
        - Vs. responding to something in the environment
- **State of knowledge:** learner's current understanding of something
    - Includes mental models, rules, frameworks, and schemas that learner currently uses to make sense of world
    - E.g., Kid might believe that all flying objects are birds -
        - This is their current state of knowledge
- **Discrete change:** happens when a learner's mind changes or reorganizes its existing understanding because of new information
    - I.e., learner moves from one mental state to another
    - E.g.,
        - Child believes all flying objects are birds, then sees an airplane and realizes that it does not have feathers, beak, etc.
            - Creates new, separate category for airplanes
            - Mental understanding changed
- Focuses on what learners know and how they come to acquire that knowledge
- Emphasizes environmental conditions and corrective feedback because these can help support learning
    - But what happens in learner's mind is important
- Instructional explanations, demonstrations, illustrative examples, and matched non-examples can help learers
    - E.g., matched non-example shows learners what something is *not*, helping understand boundaries of concept
- Focuses on the mental activities that happen before a learner gives a response
    - Recognizes processes such as:
        - Mental planning
        - Goal setting
        - Organizational strategies
- Believe environmental "cues" and instructional components alone cannot explain all of the learning that happens during instruction
    - Learners do not simply react to their environment
    - Actively process and interpret the information they receive
- Cognitivism also focuses on how learners:
    - **Attend to information**: pay attention to it
    - **Code information**: turn it into a form that can be processed and remembered
    - **Transform information**: change or reorganize it mentally
    - **Rehearse information**: repeat or practice it to help remember it
    - **Store information**: keep it in memory
    - **Retrieve information**. bring it back from memory when needed
- Learner's thoughts, beliefs, attitudes, and values can also influence the learning process

- Focus is on changing the learner by encouraging them to use appropriate learning strategies
    - I.e, learners can improve their learning by learning *how to learn* and using strategies that help them process and remember information
- Cognitivism is useful for explaining **complex forms of learning**:
    - Reasoning
    - Problem solving
    - Information processing
- Goals similar to behaviorism: both want to communicate/transfer knowledge to students as effectively and efficiently as possible
    - One way cognitivism does this is through **simplification and standardization**
        - Knowledge can be:
            - Analyzed
            - Broken down into smaller parts
            - Simplified into basic building blocks
    - E.g, information can be “sized” and “chunked” into smaller pieces so it’s easier to take in and process quickly
        - **Behaviorist:** would focus more on designing the environment to produce the desired behaviour
        - **Cognitivist:** focuses more on efficient mental processing strategies that help understand and remember the information

<u>Views on Memory</u>

- Memory is a key
- Learning happens when information is **stored in memory in an organized and meaningful way**.
    - Information is easier to remember and use when it is organized and connected to things the learner already knows.
- Teachers and instructional designers are responsible for helping learners **organize information in the best/most effective way possible**.
- Designers can use techniques such as:
    - **Advance organizers**: give learners overview or framework before introducing new information
        - They have something to connect the new information to
    - **Analogies**: explain something unfamiliar by comparing it to something familiar
    - **Hierarchical relationships**: organize info from broad/general ideas to smaller/more specific ideas
    - **Matrices**: organize info in a structured table or grid to easily see relationships and differences
- These help learners **c**onnect new information to knowledge they already have
- **Forgetting**: being unable to retrieve information from memory
    - This can happen because of:
        - **Interference**: other information gets in the way of remembering
        - **Memory loss**

        - **Missing or inadequate retrieval cues:** learner does not have the right reminder or clue needed to access the information

Views on Transfer

- Transfer depends on how info is stored in memory
    - How well information is organized and connected in memory affects whether the learner can use it somewhere else
- Transfer happens when a learner can take what they learned and apply it in a different situation or context
    - If a learner understands **rules, structures, and concepts**, vs. memorizing individual facts, more likely to:
        - Retrieve the relevant mental model from memory
        - Understand how it applies
        - Apply it to new and different problems
- Understanding is more useful for transfer than simply memorizing facts
    - If learners understand the underlying structure of knowledge, more likely to be able to use that knowledge in a new situation

Basic Principles to Instructional Design

- Many instructional strategies are used by both cognitivists and behaviorists, but for different reasons:
    - **Feedback:**
        - **Behaviorists:** use feedback to change or modify behaviour in the desired direction
        - **Cognitivists:** use knowledge of results to help the learner develop and correct accurate mental connections
            - I.e., feedback helps learner understand whether their thinking or understanding is correct and make changes if needed
    - **Learner and task analyses:**
        - **Behaviorists:** look at learners to determine:
            - Where the lesson should start
            - Which reinforcers would be most effective
            - What level the learner is currently able to perform successfully at
            - What consequences/rewards are most desired by the learner
        - **Cognitivists:** look at the learner to determine:
            - Predisposition to learning
                - What they already know, their abilities, and how prepared they are to learn
            - How instruction can be designed so that new information can be easily assimilated
                - **Assimilation**: fitting new information into an existing mental framework

                - How learning can activate, maintain, and direct the learner's learning
                - What the learner's existing mental structures are
- Assumptions/principles that have direct relevance to instructional design include:
    - Emphasis on **active involvement of the learner in the learning process**
        - Learners should actively participate rather than simply receive information
        - Can involve learner control and metacognitive training
            - **Metacognition**: thinking about and managing your own learning
        - E.g.,
            - Self-planning
            - Monitoring your own understanding/progress
            - Revising or changing your learning techniques when something is not working
    - Use of **hierarchical analysis to identify and illustrate prerequisite relationships**
        - Figuring out what a learner needs to know or be able to do first before they can learn something more advanced
        - Involve cognitive task analysis procedures, which look at the mental processes needed to complete a task
    - Emphasis on **structuring, organizing, and sequencing information to facilitate optimal processing**
        - Information should be presented in an order and structure that makes it easier for the learner to process
        - Cognitive strategies can include:
            - Outlines
            - Summaries
            - Synthesizers
            - Advance organizers
    - Creation of **learning environments that allow and encourage students to make connections with previously learned material**
        - Students should be given opportunities to connect new information with things they have already learned
        - This can include:
            - Recalling prerequisite skills
            - Using relevant examples
            - Using analogies

<u>Formulating Instruction and Assessment</u>

- Instruction should focus on making knowledge meaningful
    - Learners should be helped to organize new information and connect it to knowledge they already have in memory
    - Vs. behaviorism
        - Teachers need to arrange the environment so students respond correctly to particular stimuli

- To be effective, instruction should be based on the students' existing mental structures or schemas
    - I.e., teachers should consider what students already know when introducing something new
    - Instruction should be organized so learners can connect new information to their existing knowledge in a meaningful way
    - E.g., Instructional design textbooks often use an analogy between the familiar profession of an architect and the unfamiliar profession of **i**nstructional design
        - Student may already understand what an architect does
        - By comparing instructional design to architecture, student has something familiar to connect the new information to
        - Helps a novice **understand, organize, and remember** the major duties and functions of an instructional designer
    - Other strategies that can be used:
        - **Framing** → giving learners a clear structure or context for the information
        - **Outlining** → showing the main ideas and how they are organized
        - **Mnemonics** → memory techniques that make information easier to remember
        - **Concept mapping** → visually showing how different ideas are connected
        - **Advance organizers** → providing an overview/framework before new information is introduced
- **Task of the teacher:**
    - Understand that individuals bring different learning experiences to a learning situation
        - Previous experiences can affect their learning outcomes
    - Determine the most effective way to **organize and structure new information** so that it connects to learners' previously acquired:
        - Knowledge
        - Abilities
        - Experiences
    - Arrange **practice with feedback** so that new information can be effectively and efficiently **assimilated and/or accommodated within the learner's cognitive structure**.
        - **Assimilation** = fitting new information into an existing mental framework
        - **Accommodation** = changing an existing mental framework when new information does not fit
        - **E.g.,** Manager in the training department of a large corporation is asked to teach a new intern how to complete a cost-benefit analysis for an upcoming development project.
            - Assume the intern has **no previous experience** with cost-benefit analysis
            - Instead of teaching the procedure as something completely new, the manager can relate it to a **similar procedure that the intern already has experience with**

        - Helps manager make new procedure easier and more efficient for intern to learn and store in memoryFamiliar procedures could include:
            - How an individual allocates their monthly paycheck
            - How they decide whether to **buy something or not buy**
            - How they determine and prioritize a weekend's spending activities
        - These familiar procedures do **not exactly match cost-benefit analysis**
        - However, they are similar enough that the intern can connect the unfamiliar information to a **familiar context**
        - This makes it easier to understand the new procedure because the learner already has something in their mind to connect it to
        - It can also:
            - **Reduce processing requirements** → learner does not have to process everything as completely new information
            - **Increase the potential effectiveness of recall cues** → familiar experiences can act as reminders that help the learner retrieve the new information later

<u>Reflection</u>

- Cognitivism feels more abstract as we need to think about what is happening inside someone's mind, their "mental processes"
- However, I can relate to parts of it - assimilation for example
    - E.g., the part about learners connecting new information to knowledge they already have
    - When I learn new things that are completely unfamiliar, it is definitely faster to understand when someone compares it something I already know
- I also understand that even though behaviourism is easier to understand, just because you can see someone do something, or they provide you with the "correct" response after specific stimuli - does not necessarily mean they truly understand
- Cognitivism focuses on a deeper type of learning that wasn't well covered in behaviourism - true *understanding* vs route memorization
- When it comes to thinking about where this is applied in our world, I can see a few examples of it used in the classroom, but where this theory was applied using technology was a bit more challenging:
    - Mindmap applications: help organize and visually link new ideas to existing knowledge structures (there are many applications that help one do this, but traditional pen/paper or whiteboard is also possible)
    - Chunking: breaking a more complicated concept into smaller, more digestible pieces
        - Things like tutorials or walkthroughs online that break down a math concept into smaller steps/videos